Assignment 9

Total = 10 Marks

Multiple Choice Questions

Question 1: In the back-propagation algorithm, what is the primary purpose of the backward pass?

(a) To divide the dataset into training and validation sets
(b) To calculate the input data from the target output
(c) To propagate the prediction error backward and update network parameters
(d) None of these

[1 *Mark*]

Correct answer: (c) To propagate the prediction error backward and update network parameters

Detailed Solution:

Back-propagation first performs a forward pass to obtain the network output. The error is then propagated backward, and the weights are adjusted using gradient descent to reduce the prediction error.

Hence, the correct answer is: To propagate the prediction error backward and update network parameters.

Question 2: During neural network training, the training error continues to decrease while the cross-training error begins to increase. What does this most likely indicate?

(a) Underfitting
(b) Overfitting
(c) Successful generalization
(d) Insufficient network complexity

[1 *Mark*]

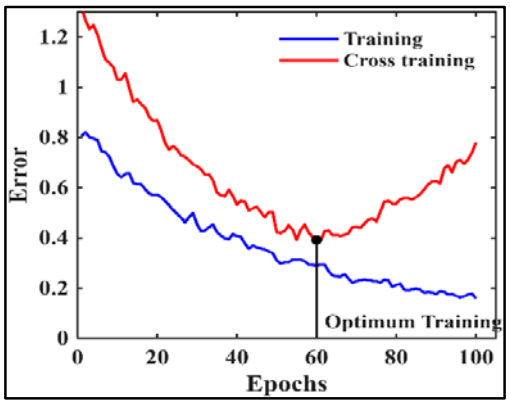
Correct answer: (b) Overfitting

Detailed Solution:

When the training error decreases but the cross-training error starts increasing, the network is becoming increasingly tuned to the training data. This indicates **overfitting**, and further training should generally be stopped.

Therefore the correct answer is overfitting.

Question 3: Consider the following figure showing the variation of training error and cross-training error with training epochs.



Which of the following statements correctly identifies the optimal training point?

(a) Training should continue until the training error becomes zero.
(b) Training should be stopped when the cross-training error reaches its minimum.
(c) Training should be stopped when both training and cross-training errors become equal.
(d) Training should continue as long as the training error decreases.

[1 *Mark*]

Correct answer: (b) Training should be stopped when the cross-training error reaches its minimum.

Detailed Solution:

Initially, both errors decrease. After the optimal training point, the training error continues to decrease, but the cross-training error begins to increase. This indicates overfitting; therefore, training should be stopped near the minimum cross-training error.

Therefore the correct answer is Training should be stopped when the cross-training error reaches its minimum.

Question 4: A 7×7 input matrix is convolved with a 3×3 filter using a stride of 2 and no padding. What is the size of the resulting output matrix?

(a) 3×3
(b) 4×4
(c) 5×5
(d) 6×6

[2 *Mark*]

Correct answer: (a) 3×3

Detailed Solution: For a two-dimensional convolution without padding,

$$Output\ size = \frac{m-f}{s} + 1 = \frac{7-3}{2} + 1 = 3$$

Where m = input width, f = filter width, s = stride.

Therefore, the correct answer is 3×3.

Question 5: Consider the following 5×5 input matrix:

2	1	3	4	2
1	2	0	3	1
3	1	2	2	4
2	3	1	4	0
1	2	3	1	2

and 3×3 filter

0	1	0
-1	0	1
0	2	0

The filter is flipped by 180∘ before performing the convolution. If the convolution is performed with stride = 1 and no padding, what is the value of the top-left element of the output matrix?

(a) -3
(b) 0
(c) 2
(d) 4

[3 *Mark*]

Correct answer: (d) 4

Detailed Solution:

Step 1: Flip the filter by 180°

The original filter

$F =$

0	1	0
-1	0	1
0	2	0

After rotating it by 180° :

$F' =$

0	2	0
1	0	-1
0	1	0

Step 2: Select the top-left 3×3 patch

From the 5×5 matrix:

$A_{patch} =$

2	1	3
1	2	0
3	1	2

Step 3: Perform element-wise multiplication

$$B = A_{patch} \times F'$$

$$B = 2 \times 0 + 1 \times 2 + 3 \times 0 + 1 \times 1 + 2 \times 0 + 0 \times (-1) + 3 \times 0 + 1 \times 1 + 2 \times 0 = 4$$

Therefore, The correct answer is 4.

Question 6: A neuron in a neural network receives an input $x = 2$. The weight associated with the input is $w = 1.5$, and the bias is $b = -1$. The neuron uses the sigmoid activation function. What is the output of neuron ?

(a) 0.29
(b) 0.47
(c) 0.88
(d) 1.32

[2 *Mark*]

Correct answer: (c) 0.88

Detailed Solution:

First, calculate the net input to the neuron:

$$z = wx + b$$

Given, $w = 1.5$, $x = 2$ and $b = -1$,

Threfore,

$$z = 1.5 \times 2 + (-1) = 2$$

For the sigmoid activation function output :

$$f(z) = \frac{1}{1 + e^{-z}}$$

Now, applying value of the z in the sigmoid activation function

$$f(2) = \frac{1}{1 + e^{-2}} = \frac{1}{1 + 0.1353} = 0.88$$

Therefore, the correct answer is 0.88.